

Patterns

Functional

-  656,250 received data and 656,250 transmitted data

Stuck at

-  33 patterns (2170 test cycles)

Transition delay

-  37 patterns (2504 test cycles)

Path delay

-  6 patterns (434 test cycles)

RS232-T1900

🔑 Trojan Description

- 🔑 Trojan trigger is a sequential comparator. Trigger input probability is $3.37e-6$ while it is externally controlled. Whenever Trojan gets triggered, its payload gains control over xmit_doneH primary output signal.

🔑 Trojan Taxonomy

- 🔑 Insertion phase: Design
- 🔑 Abstraction level: Layout-level
- 🔑 Activation mechanism: Externally conditionally triggered
- 🔑 Effects: Change functionality
- 🔑 Physical characteristics: Functional

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AND2X1 U309 (.A(sys_clk), .B(ena), .Y(sys_clk_temp));
SDFSFRX1 iDatsend_reg_1 (.CK(sys_clk_temp), .D(iRECEIVER_state_0_temp), .Q(n400), .RN(n264), .SE(test_se), .SI(iXMIT_bitCell_cntrH_0_), .SN(1'b1));
SDFSFRX1 iDatsend_reg_2 (.CK(sys_clk_temp), .D(n190_temp), .Q(n401), .RN(n266), .SE(test_se), .SI(iXMIT_state_2_), .SN(1'b1));
SDFSFRX1 iDatsend_reg_3 (.CK(sys_clk_temp), .D(uart_REC_dataH_temp), .Q(n402), .RN(1'b1), .SE(test_se), .SI(iRECEIVER_rec_datH), .SN(n266));
SDFSFRX1 iDatsend_reg_4 (.CK(sys_clk_temp), .D(iRECEIVER_rec_datSyncH_temp), .Q(n403_temp), .RN(1'b1), .SE(test_se), .SI(rec_dataH_rec[7]), .SN(n266));
AND2X1 U305 (.A(ena), .B(iRECEIVER_state_0_), .Y(iRECEIVER_state_0_temp));
AND2X1 U306 (.A(ena), .B(n190), .Y(n190_temp));
AND2X1 U307 (.A(ena), .B(uart_REC_dataH), .Y(uart_REC_dataH_temp));
AND2X1 U308 (.A(ena), .B(iRECEIVER_rec_datSyncH), .Y(iRECEIVER_rec_datSyncH_temp));
AND2X1 U304 (.A(ena), .B(n403_temp), .Y(n403));
NAND3X1 U400 (.A(iXMIT_next_state_2_), .B(iXMIT_N46), .C(ena), .Y(iXMIT_state_CTRL_2));
NAND4X1 U292 (.A(ena), .B(iXMIT_state_0_), .C(iXMIT_state_1_), .D(iXMIT_state_2_), .Y(iXMIT_state_CTRL_1));
NAND4X1 U293 (.A(ena), .B(iXMIT_N45), .C(iXMIT_N44), .D(iXMIT_N29), .Y(iXMIT_N_CTRL_1));
NAND4X1 U294 (.A(n400), .B(n401), .C(n402), .D(n403), .Y(iXMIT_N_CTRL_2));
OR4X1 U296 (.A(iXMIT_state_CTRL_2), .B(iXMIT_N_CTRL_1), .C(iXMIT_N_CTRL_2), .D(iXMIT_state_CTRL_1), .Y(iXMIT_CTRL));
AND2X1 U303 (.A(iXMIT_CTRL), .B(xmit_doneH_temp), .Y(xmit_doneH));
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RS232-T1900

Functional

Trojan activation probability	The number of Transmitted	The number of Received	Trojan activation	Payload Change	Simulation Length (No. Clock)	Length of each Receive/Transmission
3.37e-6	656 250	656 250	1	0	168 000 000	256/256

Structural

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Stuck-at unmatched
67 200	67 648	8 151	34	0	0	0

Transition delay

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Transition unmatched
87 811	88 404	9 880	88	0	0	0

Path delay

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Transition unmatched
34 573	34 777	3 278	12	0	0	0